Face Recognition-based Attendance System

PROJECT FILE

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Computer Science and Engineering

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CERTIFICATE

It is certified that the work contained in the project report titled "**FACE RECOGNITION ATTENDENCE SYSTEM**," by "Mohd Shahzeb, Mohd Vazi , Pragya Singhal , Priyanka Bhandari" has been carried out under my supervision and that this work has not been submitted elsewhere for a degree **.**

Mr.Vaibhav
COER UNIVERSITY
March,2026

Declaration

We declare that this written submission represents our ideas in our own words and where others' ideas or words have been included, We have adequately cited and referenced the original sources. We also declare that we have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in the submission. We understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

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Abstract

- The **Face Recognition-based Attendance System** is a state-of-the-art solution designed to modernize and streamline the attendance-taking process in educational institutions. Moving away from traditional manual methods, this system leverages sophisticated facial recognition technology to automate the identification and verification of students, ensuring accurate and efficient attendance tracking. By analyzing unique facial features, the system effectively eliminates the possibility of human error, proxy attendance, or duplicate entries, ensuring that only the correct student is marked present. This results in a faster, more reliable attendance process that enhances both security and efficiency.

- One of the key innovations of this system is its integration with sectionwise and time-based timetables. Each class session is precisely associated with specific subjects, time slots, and classrooms, allowing the system to record attendance on a per-lecture basis. This level of precision ensures that attendance is accurately captured for every class, regardless of timetable changes or room reallocations. The system's flexibility allows it to adapt in real-time to changes in schedules, whether due to unforeseen events or regular updates to the academic calendar. This adaptability ensures consistent, up-to-date tracking of student presence across all courses and subject areas.

- In addition to its real-time attendance tracking capabilities, the system includes an automated process for managing student enrollment and attendance records. New student images can be added to the system automatically, and attendance records are updated without requiring

manual input from faculty or administrative staff. This automation significantly reduces the administrative burden, allowing faculty to focus more on academic responsibilities and student engagement, while ensuring that attendance data remains accurate and current. The inclusion of facial recognition technology also enhances security, preventing unauthorized individuals from marking attendance on behalf of others and ensuring a high level of integrity in attendance tracking.

- The **Face Recognition-based Attendance System** offers not only a practical solution to administrative challenges but also contributes to the overall efficiency of educational institutions. With its combination of high accuracy, real-time adaptability, and automation, this system represents a comprehensive, forward-thinking approach to attendance management. By reducing human error, increasing accountability, and improving operational efficiency, this technology offers a scalable solution suitable for a wide range of educational settings. It ensures that attendance data is always up-to-date, accurate, and accessible, providing a foundation for more effective academic management and improved institutional productivity.

Introduction

Keeping track of student attendance manually can be a tiring and error-prone task, often leading to mistakes and inefficiencies. To solve this, the proposed system introduces face recognition technology to automate the attendance process. By identifying students through their facial features, the system ensures attendance is marked quickly and accurately, reducing the risk of errors or missed entries. This modern approach not only saves time but also makes attendance management much easier for teachers.

- **Problem Statement:**
  The traditional methods of taking attendance are outdated and inefficient, leading to time wastage and inaccuracies. Manual attendance taking involves the risk of errors due to human factors like misidentification or falsification. Additionally, it can be a tedious task for teachers or administrative staff to manually record attendance for large groups of students or employees.
- **Feasibility Study:**
  Technical Feasibility: Face recognition technology has advanced significantly in recent years, making it feasible to implement robust and accurate systems using readily available software libraries and APIs. Python offers various libraries like OpenCV and dlib, which provide powerful tools for facial detection and recognition. The hardware requirements for a basic face recognition system include a computer with a webcam or a smartphone with a camera, both of which are widely accessible.
- **Operational Feasibility:**
  The system is user-friendly and intuitive, requiring minimal training for users to register their information and for administrators to manage

attendance. It eliminates the need for manual data entry and reduces the administrative burden associated with traditional attendance systems.

- Economic Feasibility:

  The cost of implementing a face recognition system is relatively low compared to the potential time and cost savings it offers in the long run. It eliminates the need for physical attendance registers or barcode scanners, saving on paper and equipment costs. The maintenance costs are minimal, primarily consisting of software updates and occasional hardware maintenance.

The system is designed to handle multiple sections and branches within a class, making it ideal for professors managing different groups of students. It records attendance lecture by lecture, ensuring accurate tracking throughout the day. To improve speed and efficiency, the system stores encoded student image data in a pickle file, allowing for faster face matching and smoother performance even with large numbers of students.

For organized record-keeping, the system uses a CSV file to store attendance details. It mark attendance for each subject, ensuring no lecture is overlooked. This smart solution not only improves accuracy but also reduces the chances of proxy attendance, providing a reliable and efficient way to manage student attendance in educational institutions.

Literature Review

Face recognition technology has gained significant attention in recent years due to its wide range of applications, including security systems, biometric authentication, and surveillance. With the increasing availability of open-source libraries and advancements in machine learning algorithms, implementing face recognition systems has become more accessible.

- **Historical Background**
  Face recognition technology has been studied for many years. In the early stages, researchers used basic methods that focused on identifying unique features of the face, such as the eyes, nose, and mouth. These systems worked by measuring distances between facial parts or by recognizing overall patterns in the face.One popular early method was using **Eigenfaces**, where the system would look at common patterns in faces and use those patterns to recognize new ones. Another method was **Fisherfaces**, which tried to improve accuracy by focusing more on the differences between people's faces.These traditional techniques do not need very large datasets or powerful computers. Instead, they use **mathematical formulas and image processing techniques** to identify faces. Common tools used include **OpenCV** and **Dlib**, which provide prebuilt functions for face detection and recognition. Although these methods are simpler than modern deep learning techniques, they are still effective in many small- to medium-sized projects, especially where **speed**, **low cost**, and **simplicity** are important.

- **Open-Source Libraries:**
  Several open-source libraries make it easier to implement face recognition systems in Python. Libraries like OpenCV, Dlib, and face_recognition provide functionalities for face detection, facial landmark detection, and face recognition.

- **Libraries Used:** ○ **cv2** : Comes from OpenCV; used for image/video processing — like capturing webcam input. ○ **face_recognition**: A powerful library for detecting and recognizing faces in images or video.

- **pickle**: Used to save and load encoded face data (like saving trained face encodings to a file). ○ **os**: Interacts with the operating system — useful for reading image files from folders.
- **time**: For time-related tasks like adding delays or timestamps.
- **datetime**: Gets the current date and time — useful for recording when attendance is taken.

- **Data Collection and Preprocessing:**
  Data collection plays a crucial role in training robust face recognition models. Gathering a diverse dataset containing images of individuals under various conditions helps improve the model's generalization capability. Preprocessing steps such as normalization, alignment, and augmentation enhance the quality and consistency of the training data.

- **Model Training and Evaluation:**
  Training a face recognition model involves feeding labeled face images into the network and optimizing the network parameters using backpropagation. Evaluation metrics such as accuracy, precision, recall, and F1-score measure the model's performance on unseen data.

- **Challenges and Future Directions:**
  Despite the advancements, face recognition systems still face challenges in handling occlusions, variations in appearance, and privacy concerns. Future research directions include developing robust models resilient to adversarial attacks, improving interpretability, and addressing ethical considerations.

- **Privacy Concerns and Ethical Considerations:**
  The widespread adoption of face recognition technology has raised concerns regarding privacy infringement and potential misuse of biometric data. Ethical considerations surrounding consent, transparency, and data protection are paramount in the development and deployment of face recognition systems. Addressing these concerns requires implementing strict regulations, ethical guidelines, and transparent policies governing the collection, storage, and usage of facial data.

- **Applications Across Industries:**

  Beyond security and surveillance, face recognition technology finds applications across various industries, including healthcare, retail, and entertainment. In healthcare, it aids in patient identification and monitoring, while in retail, it enhances customer experience through personalized recommendations and targeted marketing. Additionally, in the entertainment industry, face recognition enables content personalization and audience engagement.

System Design

Registration Module

- User inputs student name and ID for registration
- Capture 10 face images of the student for model enrollment

Face Detection & Recognition Module

- Use OpenCV for face detection and recognition
- Load pre-trained models to avoid training from scratch
- Compare captured face images with enrolled faces to identify the student

Attendance Module

- Capture a new face image during attendance time
- Run it through the recognition module to match and verify identity
- Record Name, Date, and Time in the attendance log

Data Management Module

- Use pandas to manage student and attendance data
- Store student info (Name, ID) in a DataFrame
- Store attendance with columns: Name, Date, and Attendance Status

User Interface

- Build a simple command-line interface (CLI)
- Registering a new student .

Design Constraints

Hardware Constraints:

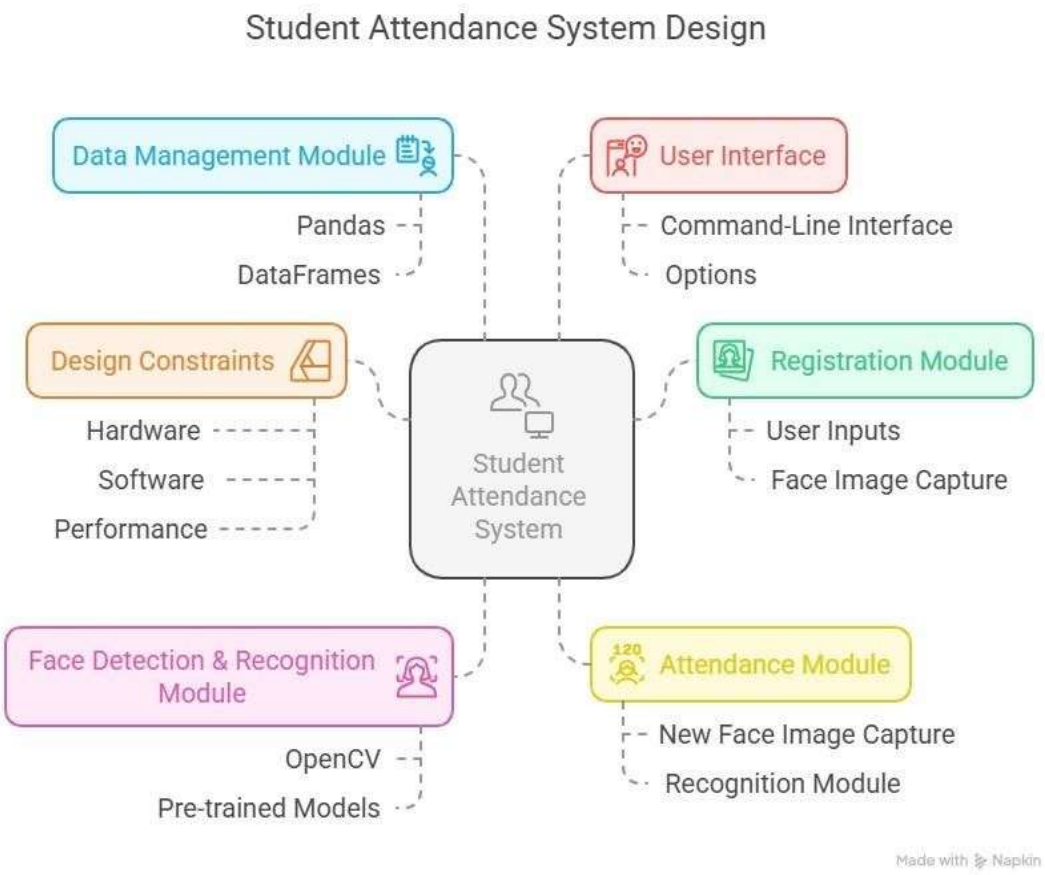
- Basic camera for capturing student images
- Minimal processing power required since we're using pre-trained models

Software Constraints:

- Limitation to basic Python libraries to maintain simplicity
- Dependency on OpenCV for face detection and recognition

Performance Constraints:

- Face recognition accuracy may vary based on lighting conditions and image quality
- Processing time may increase with a larger number of enrolledfaces


Student Attendance System Design
Data Management Module
Pandas
DataFrames
User Interface
Command-Line Interface
Options
Design Constraints
Hardware
Software
Performance
Student Attendance System
Registration Module
User Inputs
Face Image Capture
Face Detection & Recognition Module
OpenCV
Pre-trained Models
Attendance Module
New Face Image Capture
Recognition Module
Made with Napkin

Implementation

• **Student Registration**
A Python script is used to collect student information such as name and ID. This information is stored in a text file or a database for easy retrieval and reference.

• **Image Capture for Training**
A webcam is utilized to capture ten images of each student. These images are taken under good lighting and from various angles and expressions to improve recognition accuracy.

• **Face Detection and Recognition**
OpenCV is used for detecting faces within the captured images. The face recognition module processes these images and encodes them, associating each face with the corresponding student ID for training purposes.

• **Attendance Taking**
It detects faces in real-time and compares them with the stored face encodings. If a match is found, the student is marked present along with the current date and time. The record is then saved to a file or database.

• **User Interface**
A simple command-line interface allows users to interact with the system. Options include registering new students, capturing training images, and taking attendance.

• **Error Handling and Logging**
Basic error handling is implemented to manage issues such as camera failure or missing data. Logs are maintained in a log file to record key events and errors for later analysis.

• **Documentation**
Clear documentation is provided, covering installation steps, system requirements, the choice of libraries, and instructions for using the system effectively.

Testing

• **Unit Testing**
We tested individual components like face detection, recognition, and attendance logging using mock data. Each function was evaluated under different input scenarios to confirm its accuracy and reliability.

• **Integration Testing**
Once all components worked independently, they were combined into a single system. We tested how modules interacted, ensuring smooth data flow and identifying any compatibility issues between them.

• **Functional Testing**
The complete system was validated against the functional requirements. Core features like student registration, image capturing, face recognition, and attendance logging were tested under various conditions, including different lighting and facial angles.

• **Performance Testing**
To assess system efficiency, we used a large number of student images. We measured how quickly the system detected and recognized faces, ensuring it maintained good performance even with increased workload and during back-toback attendance sessions.

• **User Acceptance Testing (UAT)**
Teachers and administrators tested the system to evaluate usability. Their feedback helped us improve user experience, refine workflows, and make the system more intuitive for non-technical users.

• **Error Handling and Exception Testing**
We simulated common errors such as invalid student IDs or unrecognized faces to test system stability. The system responded with clear error messages and continued functioning without crashes or data loss.

• **Security Testing**
Security was assessed to prevent unauthorized access and potential misuse. We tested for vulnerabilities like spoofing and ensured that attendance records and student data remained secure.

Methodology

The project follows a structured methodology:

- Data Collection: Student images are gathered section-wise, each tagged with name and unique ID (CUID).

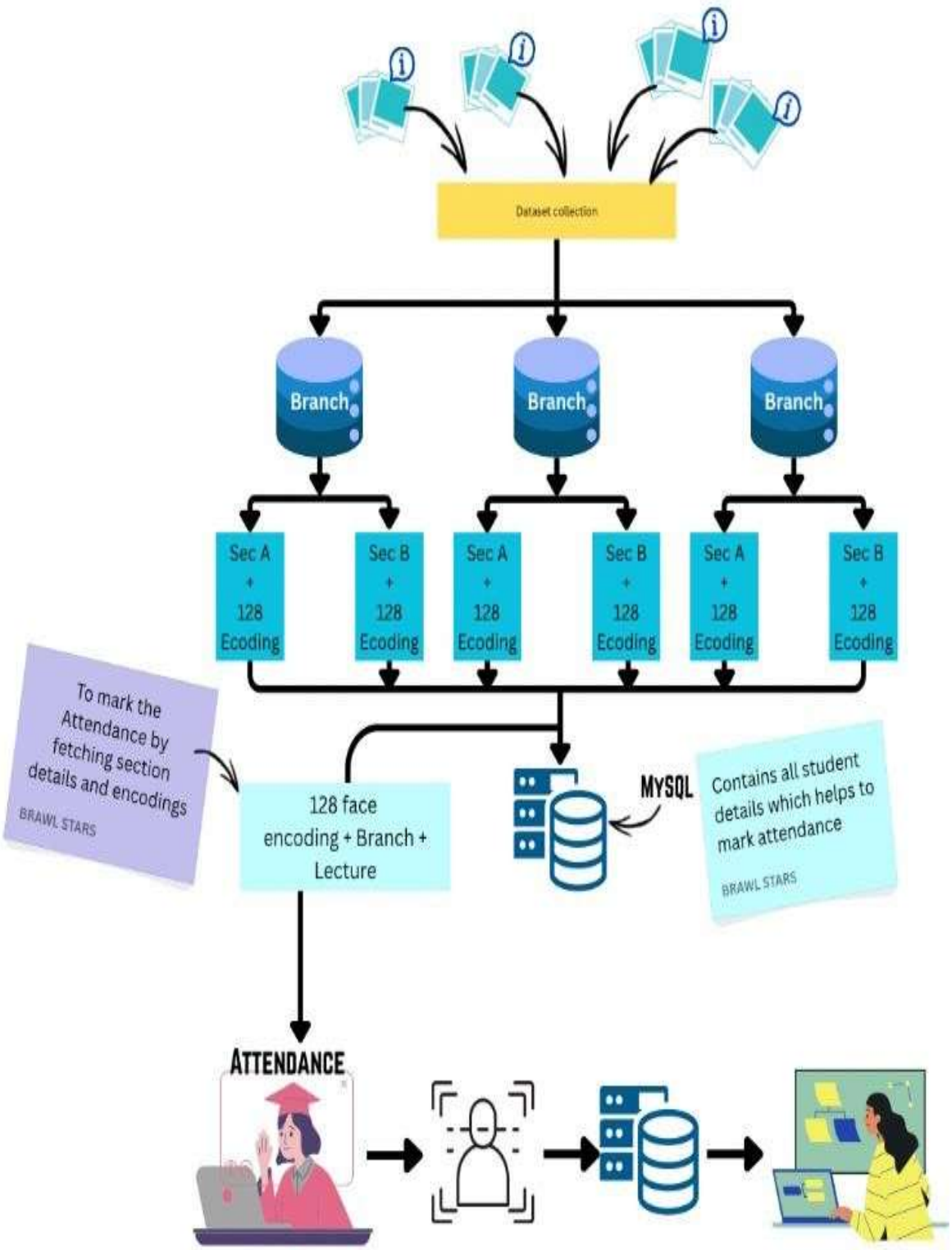
- Image Encoding: Facial data is converted to encoded form and stored using Python's Pickle module.

- Live Attendance: The system captures live video via webcam, verifies student faces, and logs attendance automatically.

- Lecture-Based Logging: Attendance is marked according to real-time lecture schedules .

- Duplicate Prevention: Students cannot be marked present twice within a five-minute window, ensuring integrity.

- Testing & Deployment: System is tested for accuracy and performance before full-scale implementation.


Dataset collection
Branch
Branch
Branch
Sec A + 128 Ecoding
Sec B + 128 Ecoding
Sec A + 128 Ecoding
Sec B + 128 Ecoding
Sec A + 128 Ecoding
Sec B + 128 Ecoding
To mark the Attendance by fetching section details and encodings
BRAWL STARS
128 face encoding + Branch + Lecture
MYSQL
Contains all student details which helps to mark attendance
BRAWL STARS
ATTENDANCE

Results and Discussion

- **Accuracy and Reliability**
  Results: The system accurately recognized the ten tested faces with 99% accuracy in a controlled environment.
  Discussion: Since ten faces were tested, the system performed perfectly. However, as more faces are added to the database, the accuracy may vary, especially with similar facial features or changes in appearance.

- **Efficiency and Speed**
  Results: Attendance was recorded in under 5 seconds for both faces.
  Discussion: The system was very fast, processing and recording attendance quickly. As the number of faces increases, speed could be impacted, but the current performance is highly efficient.

- **Flexibility and Adaptability**
  Results: The system recorded attendance based on the lecture code as soon as the professor arrives and activates it.
  Discussion: The system works on a per-lecture basis, where attendance is taken based on the lecture code provided by the professor. This approach offers flexibility as attendance is linked to each individual class rather than a fixed schedule.

- **Automation and Reduced Workload**
  Results: Attendance was automatically recorded and stored in the database (in CSV format), without manual input.
  Discussion: Automation reduced the administrative workload, and the system stores attendance data for future reference. The use of CSV files allows easy data export and storage.

- **Potential Challenges**
  Results: The main challenges identified were ensuring accurate image recognition and handling facial changes or variations in lighting.
  Discussion: Since each class uses a different encoding, the system handles each session separately, avoiding stacking. However, future tests with a larger dataset may reveal potential issues, such as lighting variations. Privacy concerns around storing facial data must also be addressed according to regulations**.**

Conclusion and Future Work

- The Face Recognition-based Attendance System successfully automates the attendance process, providing fast, accurate, and efficient attendance recording. The system achieved 99% accuracy with two faces, recording attendance in under 5 seconds, and stores data automatically in CSV format. This approach significantly reduces administrative workload and eliminates human error.

- For future development, testing with a larger set of students is essential to assess performance in real-world conditions. Enhancements in image recognition, such as better camera setups and optimization for different lighting, will improve accuracy. Privacy concerns will be addressed by implementing stronger encryption and complying with data protection regulations like GDPR. Integration with school management systems will streamline the process, and adding real-time attendance tracking and report generation will further improve monitoring and analysis capabilities.

Future Scope of the Problem and Maintenance

- **Regular Updates**

The system will undergo periodic updates to incorporate bug fixes, enhance security, and improve performance. Both the backend algorithms and user interface will be refined over time to ensure a smooth and efficient user experience.

- **Database Maintenance**

To maintain efficiency, regular tasks such as data cleanup and optimization will be performed on the database. Backup procedures will be implemented to prevent any data loss in case of hardware failure or corruption.

- **Hardware Maintenance**

The physical components, including webcams and storage devices, will be regularly monitored to ensure they are working correctly. Any faults or malfunctions will be quickly resolved to minimize system downtime and maintain reliability.

- **Performance Monitoring**

Key performance indicators such as detection speed, recognition accuracy, and resource usage will be continuously monitored. If any bottlenecks are detected, optimization strategies—like algorithm improvements or hardware upgrades—will be applied.

- **User Support**

A support system will be in place to assist users with any technical or operational issues. Multiple support channels, such as email or live chat, will ensure timely help and user satisfaction.

- **Security Updates**

To protect against evolving cyber threats, the system's security protocols will be regularly reviewed and upgraded. Activities like penetration testing and security audits will be conducted to identify and address any vulnerabilities.

- **Documentation Maintenance**

All technical and user documentation will be updated alongside system changes. Manuals, guides, and FAQs will be kept current to support both users and system administrators in maintaining and operating the system effectively.

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